

Microeconomics II – Class 7

Axioms of revealed preferences



Class summary

- * Revealed preferences
 - * Assumptions
 - * Principle of revealed preferences
 - * Properties of preferences
 - * How to “trap” the indifference curve
 - * Weak Axiom of Revealed Preferences (WARP)
- * Readings: Varian (chap. 7 and 3.1-3.3)



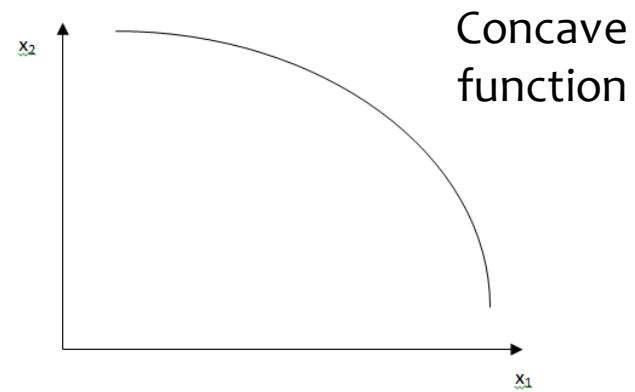
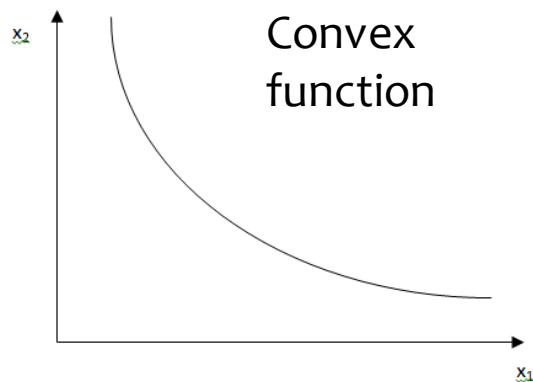
Revealed preferences

- * 2 approaches to microeconomic analysis:
 - * Preference-based approach
 - * Assumes the preferences of rational consumers and infers their choices through deductive reasoning
 - * Standard approach in Microeconomics I
 - * Choice-based approach
 - * Observes consumers' choices, which are assumed to be consistent, and infers their preferences through inductive reasoning
 - * This is the topic of this class



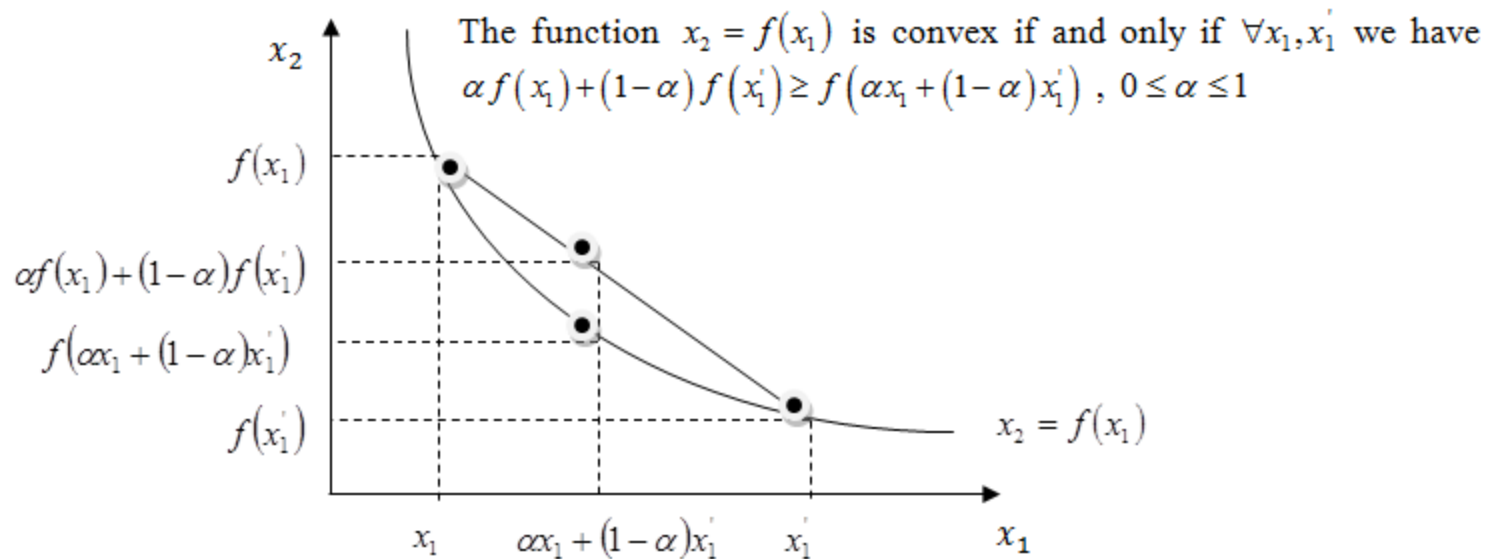
Assumptions

- * Preferences are:
 - * Stable;
 - * Monotonous: $y_1 > x_1 \Rightarrow U(y_1, x_2) > U(x_1, x_2)$
 - * Convex (implying convex indifference curves).



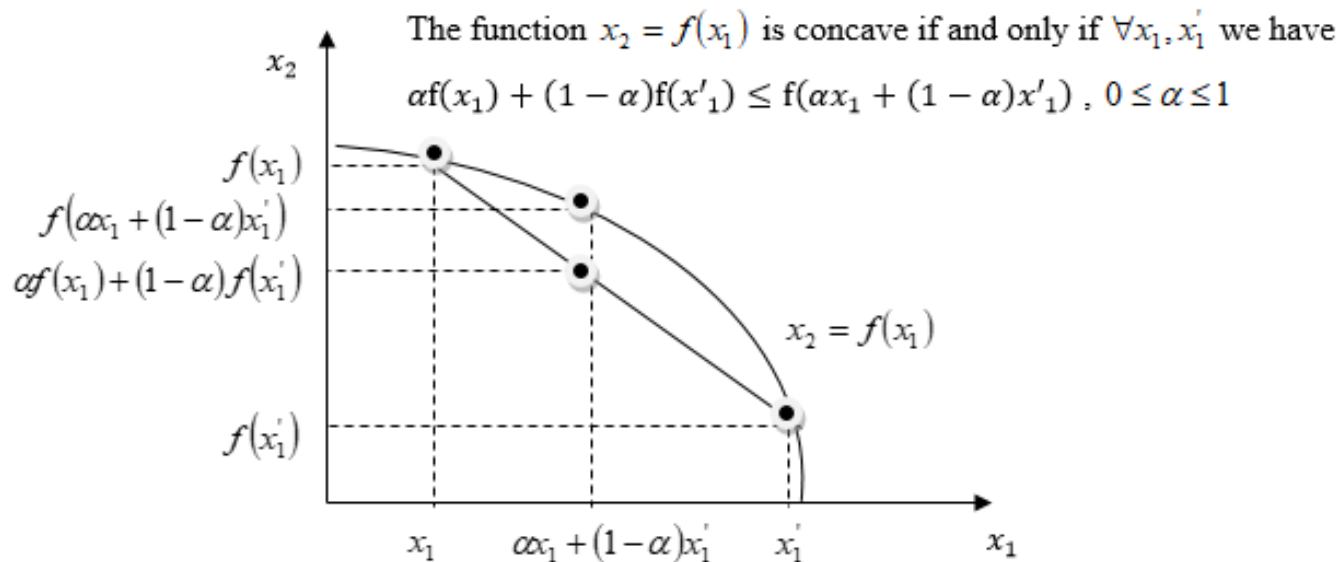
Assumptions

* Convexity – formal definition



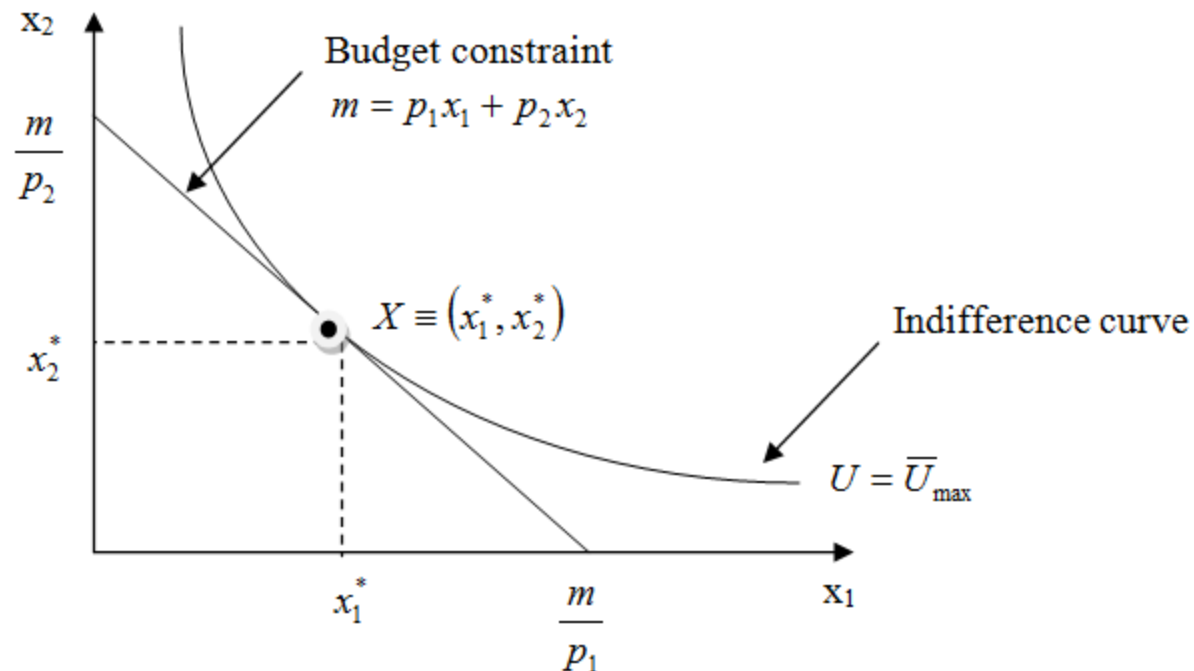
Assumptions

* Concavity – formal definition



Assumptions

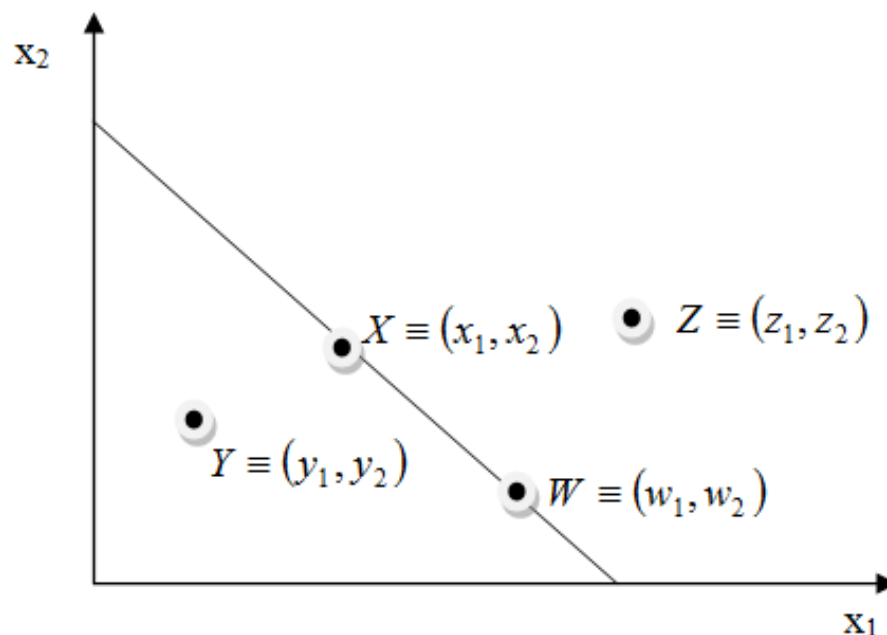
* Convex preferences $\Rightarrow$ A single optimal consumption bundle



Principle of revealed preferences

- * Knowing that the consumption bundle $X \equiv (x_1, x_2)$ was chosen, what can we say about the following alternatives?

- * X vs Y ?
- * X vs W ?
- * X vs Z ?



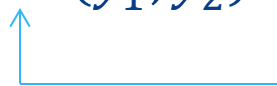
Answer: $X \succ Y \wedge X \succ W$

The revealed preference relations are strict. Recall that the convexity assumption implies that the optimal point is unique (we cannot have $X \sim W$).



Principle of revealed preferences

- * Principle of revealed preferences (direct revelation)
 - * If (x_1, x_2) is the chosen/observed consumption bundle, (p_1, p_2) are the prices of each of the goods, m is the consumer's income, and (y_1, y_2) is another bundle ($\neq (x_1, x_2)$) such that
$$m = p_1 \times x_1 + p_2 \times x_2 \geq p_1 \times y_1 + p_2 \times y_2$$
then $(x_1, x_2) \succ (y_1, y_2)$



Strict preference because we assumed convexity.



Principle of revealed preferences

- * Symbols for preference relations

- $\succ$ strict preference (it is strictly preferred to)

- $\sim$ indifference (it is indifferent to)

- $\succeq$ weak preference (is it weakly preferred to; it is at least as good as)



Properties of preferences

- * Completeness

(any two consumption bundles can be compared)

$\forall X, Y$ we have $X \succeq Y \wedge Y \succeq X$

- * Reflexivity

- * Any consumption bundle is at least as good as itself (apparently obvious but necessary to guarantee consistent rational behavior)

$X \succeq X$

- * Transitivity

$X \succeq Y \wedge Y \succeq Z \Rightarrow X \succeq Z$



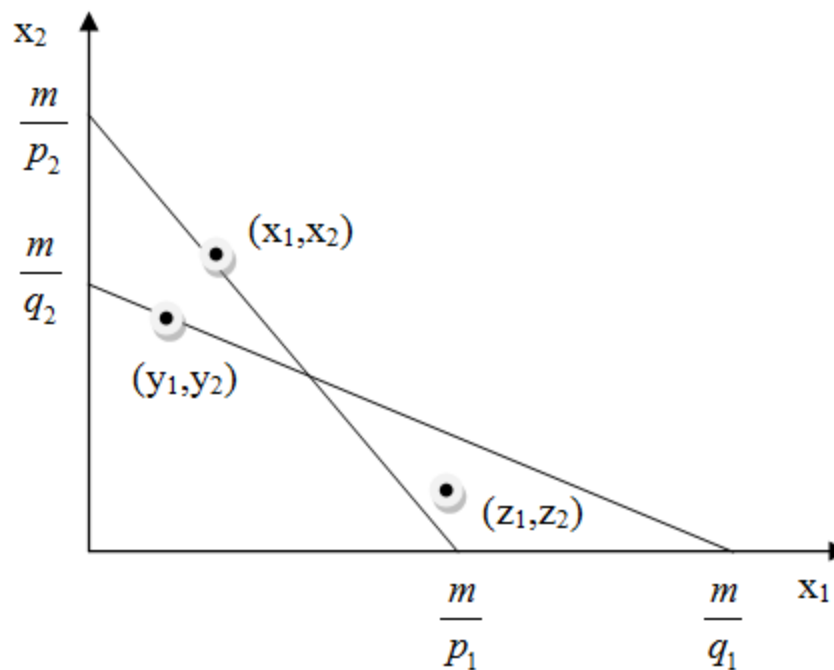
Principle of revealed preferences

- * Indirect preference revelation

- * We know that

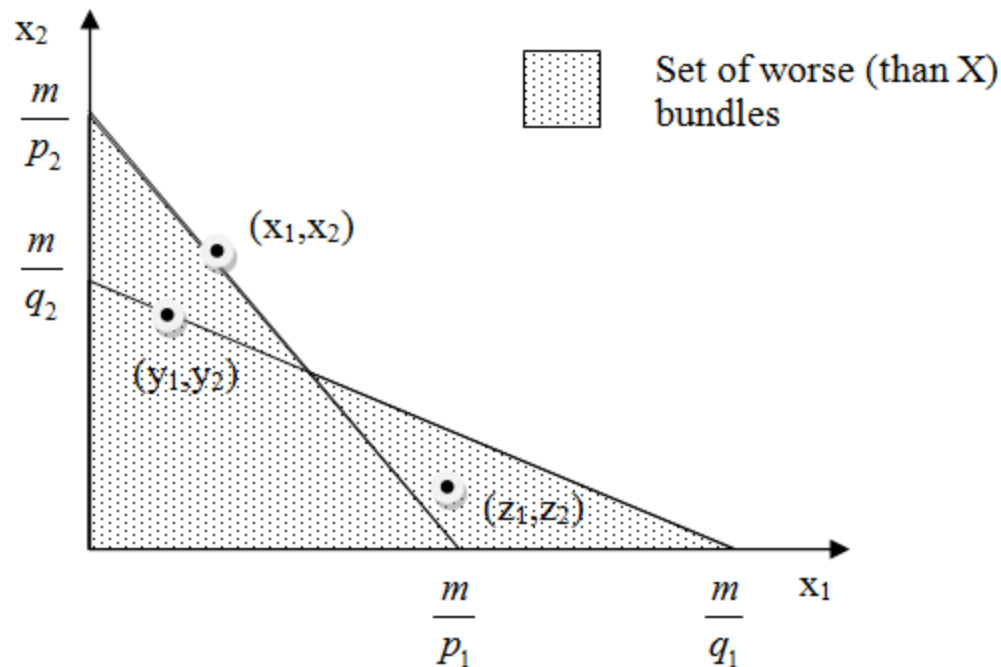
$$X \succ Y \wedge Y \succ Z$$

- * therefore $X \succ Z$



How to “trap” the indifference curve

- * Set of worse bundles



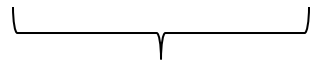
How to “trap” the indifference curve

- * Set of better bundles

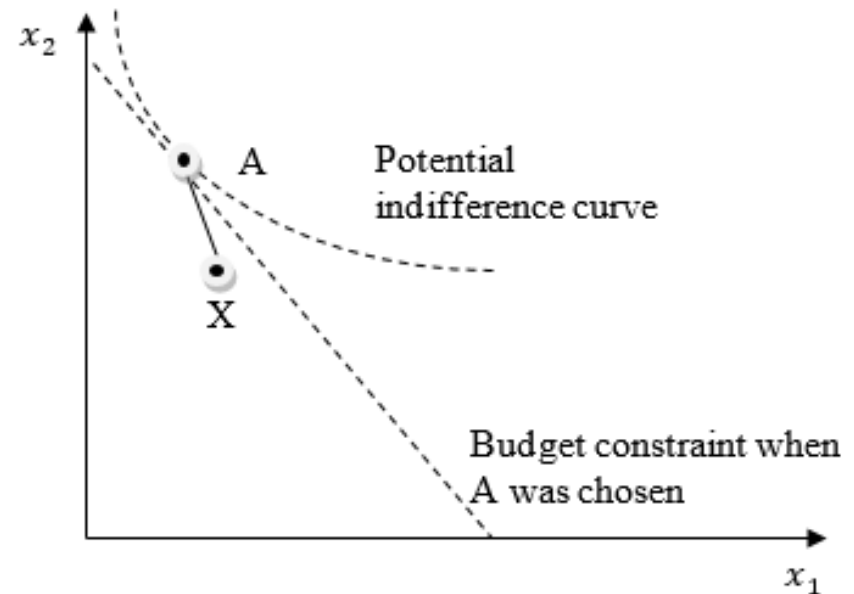
$A \succ X \wedge \text{convexity}$

$\Rightarrow$

$$tA + (1 - t)X \succ X, \quad 0 < t < 1$$



Bundles in the line segment $]AX[$



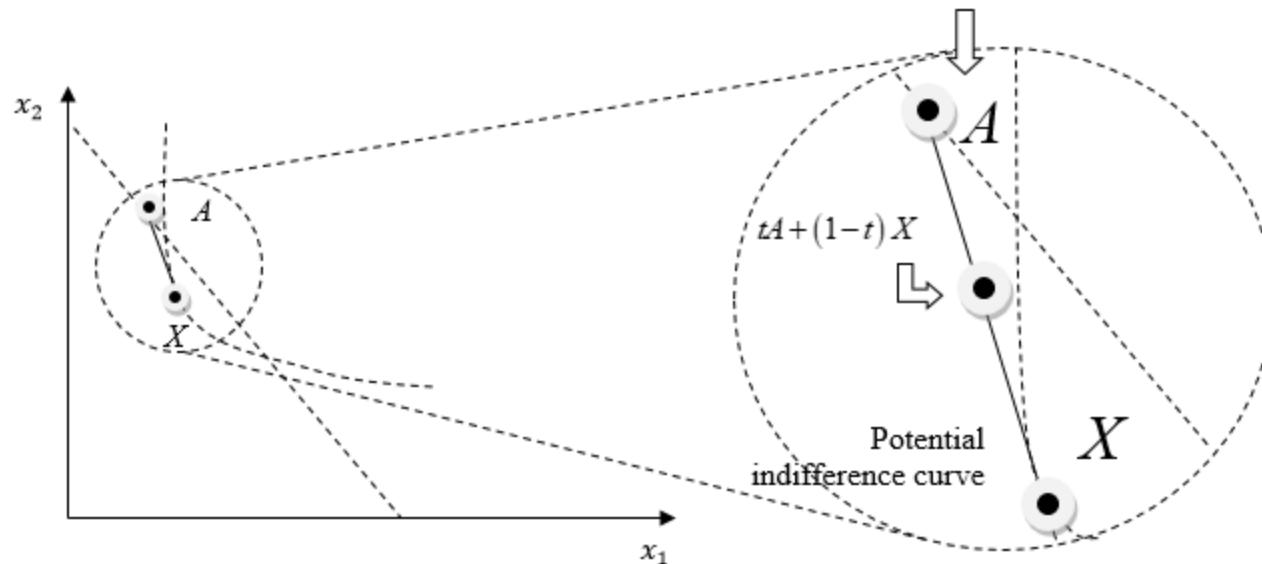
How to “trap” the indifference curve

- * Set of better bundles

$$A \succ X \wedge \text{convexity} \Rightarrow tA + (1-t)X \succ X, \quad 0 < t \leq 1$$

Proof by contradiction: Suppose $X \succ tA + (1 - t)X$, $0 < t \leq 1$

$X \succ A$, which contradicts the fact that $A \succ X$



How to “trap” the indifference curve

* Set of better bundles

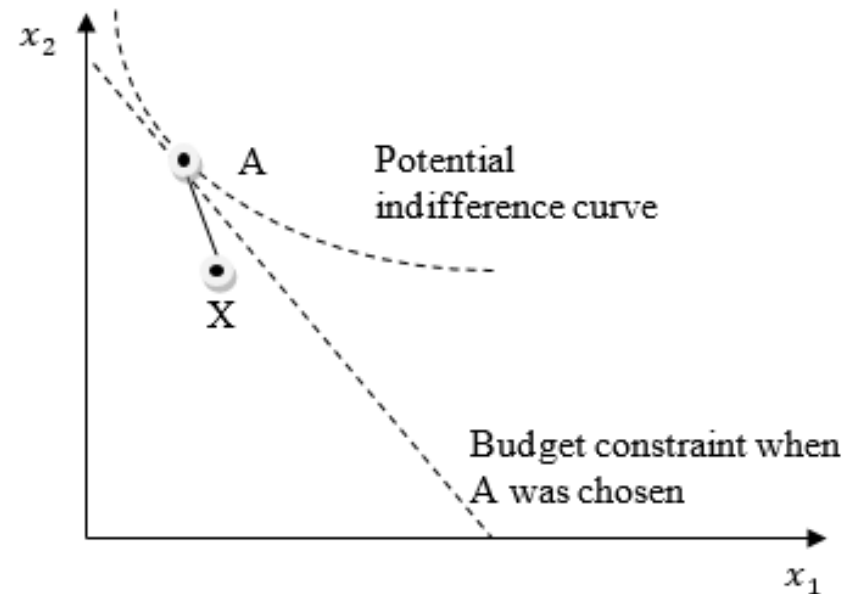
If, assuming convexity, when $X \succ A$ we cannot have either

$$X \sim tA + (1 - t)X$$

$$X \succ tA + (1 - t)X$$

it must be the case that

$$tA + (1 - t)X \succ X$$

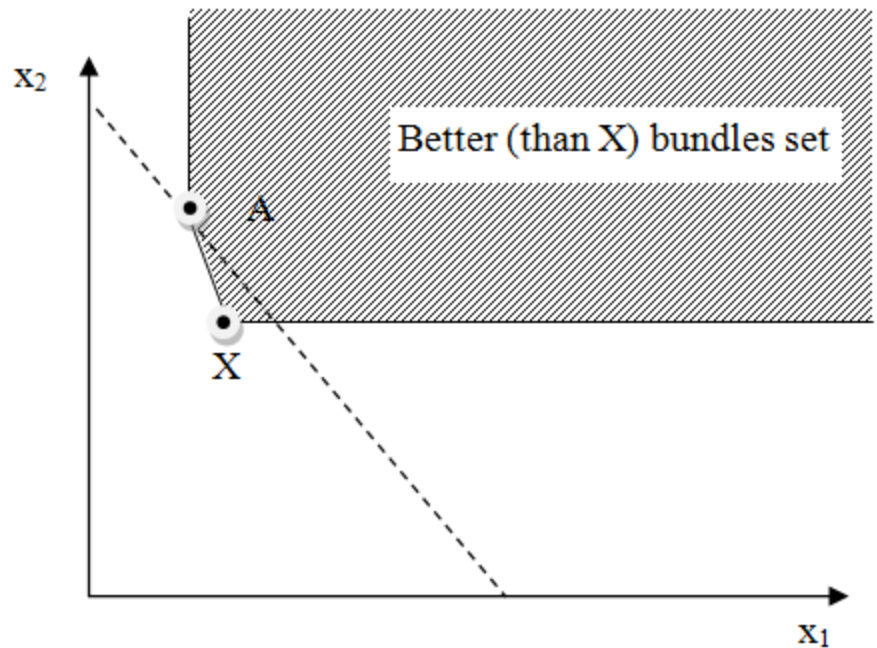


How to “trap” the indifference curve

- * Set of better bundles

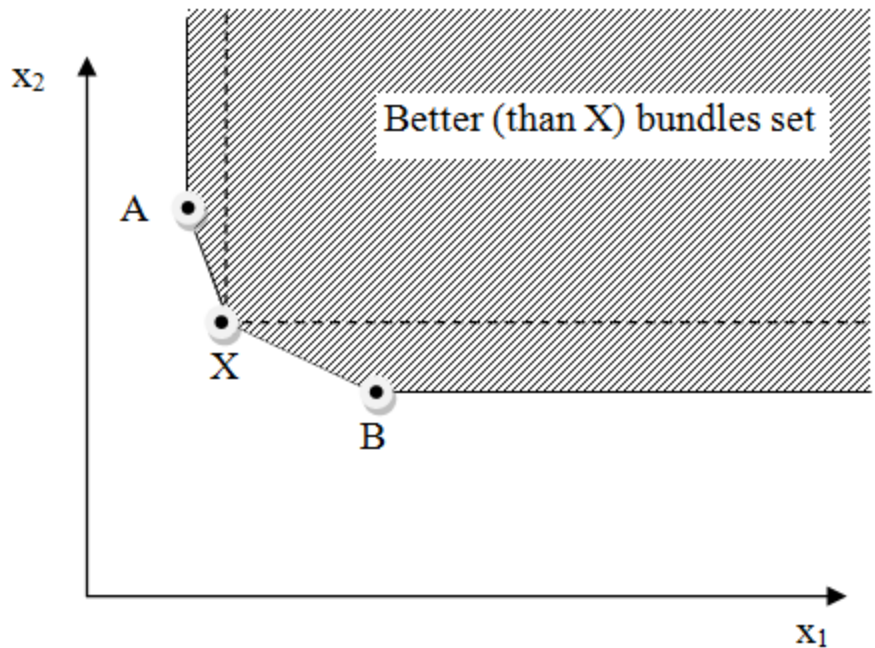
- * Considering the monotonicity and transitivity properties of preferences

$$y_1 > x_1 \Rightarrow U(y_1, x_2) > U(x_1, x_2)$$



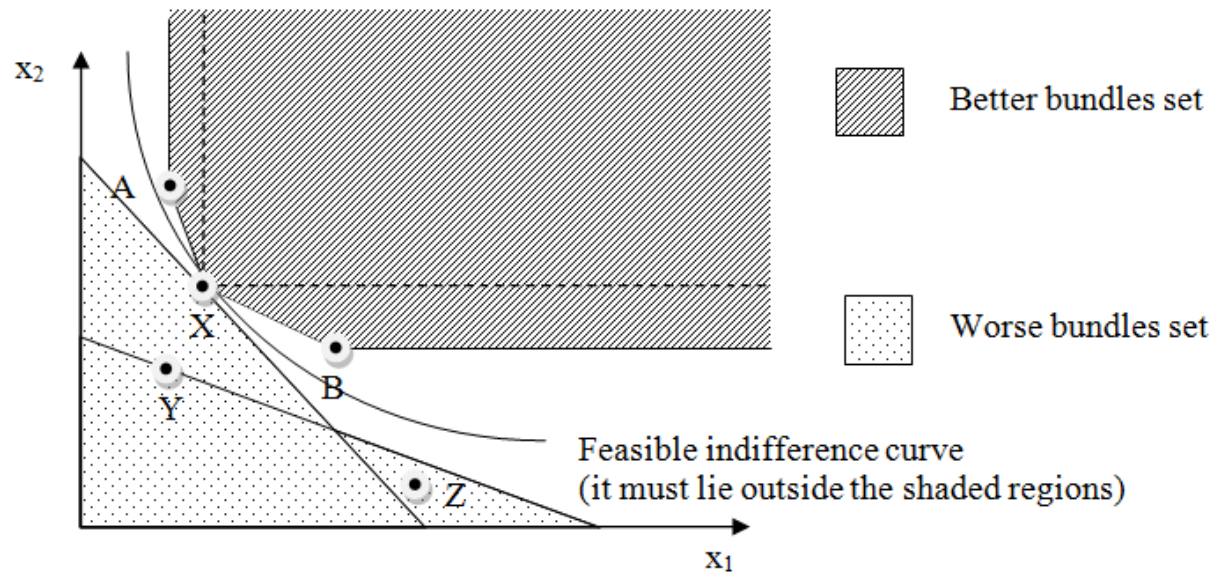
How to “trap” the indifference curve

- * Set of better bundles
 - * With more observations, the set can be enlarged / better defined



How to “trap” the indifference curve

- * Combining both the set of worse and better bundles (than X) we can limit the region where the indifference curve may lie.



Weak Axiom of Revealed Preferences (WARP)

If $(x_1, x_2) \neq (y_1, y_2) \wedge (x_1, x_2) \succ (y_1, y_2)$

It cannot be the case that $(y_1, y_2) \succ (x_1, x_2)$

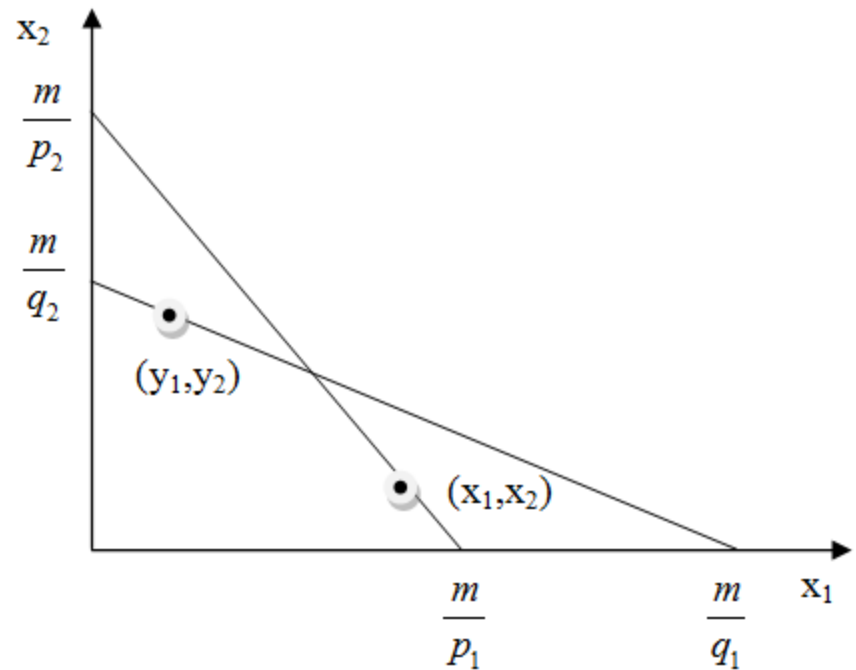
i.e.,

If (x_1, x_2) is chosen at the prices (p_1, p_2) and (y_1, y_2) is chosen at the prices (q_1, q_2) , then if $p_1x_1 + p_2x_2 \geq p_1y_1 + p_2y_2$, i.e., Y is affordable, it must not be the case that $q_1y_1 + q_2y_2 \geq q_1x_1 + q_2x_2$, i.e., X could not be affordable at the new prices without violating WARP.



Weak Axiom of Revealed Preferences (WARP)

- * Example of violation of the Weak Axiom of Revealed Preferences



Weak Axiom of Revealed Preferences (WARP)

- * Does the situation depicted to the right violate the Weak Axiom of Revealed Preferences?

